TakaID: A Comparative Study of Lightweight and Deep Learning Architectures for Automated Bangladeshi Banknote Classification

***Abstract*—Automated currency recognition is a cornerstone of modern financial technology and aids visually impaired individuals in daily transactions. This paper presents "TakaID," a robust system designed to classify Bangladeshi banknotes using deep Convolutional Neural Networks (CNNs). While digital transactions are rising, physical currency remains the dominant medium of exchange in Bangladesh. This study addresses critical challenges such as inter-class similarity, varying lighting conditions, and physical degradation of notes. We conduct a comprehensive comparative analysis of three architectures: a custom-designed lightweight CNN, MobileNetV2, and ResNet50. Our experiments, conducted on a dataset of 10,000 augmented images, demonstrate that MobileNetV2 achieves state-of-the-art performance with 99.11% test accuracy. Conversely, the Custom CNN offers a competitive 98% accuracy with significantly lower computational complexity, validating its potential for low-end devices. We further analyze the trade-offs between model size and accuracy, providing a benchmark for deploying currency recognition systems on resource-constrained mobile devices.**

***Index Terms*—Banknote Recognition, Deep Learning, MobileNetV2, ResNet50, CNN, Transfer Learning, Assistive Technology.**

I. Introduction

The rapid advancement of computer vision has enabled significant strides in assistive technologies. For the visually impaired, identifying paper currency is a daily challenge, particularly in developing economies like Bangladesh where cash is the primary mode of transaction. According to recent surveys, a significant portion of the Bangladeshi population still relies on physical cash for daily necessities. Visually impaired individuals often depend on the tactile feel of notes or the assistance of others, which exposes them to risks of fraud and financial loss [1].

Bangladeshi banknotes (Taka) pose unique difficulties for computer vision systems. Similarities in size and color exist across different denominations; for instance, the 500 BDT and 100 BDT notes share spectral features that can be confusing under artificial lighting. Furthermore, the circulation of soiled, torn, or written-on notes ("murky" currency) adds noise to the classification problem. Recent efforts have focused on identifying fraudulence in these scenarios [2].

Existing solutions often rely on traditional image processing techniques that are sensitive to lighting and background clutter. This research proposes **TakaID**, a deep learning-based solution that leverages the power of Convolutional Neural Networks (CNNs) to robustly identify denominations. We investigate three distinct approaches:

1) A **Custom CNN** designed from scratch for maximum efficiency.
2) **ResNet50**, a deep residual network, evaluated via feature extraction to test the limits of generic features.
3) **MobileNetV2**, a lightweight architecture optimized for mobile inference, evaluated via fine-tuning.

The contributions of this paper are as follows:

- Development of a diverse dataset of 10,000+ augmented images covering all 9 denominations of Bangladeshi Taka.
- A rigorous comparative analysis of three architectures showing MobileNetV2's superiority (99.11% accuracy).
- Detailed error analysis identifying the specific confusion between 50 and 100 Taka notes.
- A feasibility study for mobile deployment, highlighting the trade-off between model parameters and accuracy.

II. Related Work

Currency recognition has been an active area of research for decades. Early approaches relied heavily on hand-crafted feature extraction, but the advent of Convolutional Neural Networks (CNNs) revolutionized this field. **Zheng et al. [5]** and **Islam et al. [6]** were among the first to successfully apply deep learning architectures to Bangladeshi banknotes, establishing the feasibility of automated recognition. Building on this, Tasnim et al. [3] proposed a custom CNN approach achieving high accuracy but with limited testing on physically degraded notes. Similarly, Shoumik et al. [4] explored lightweight architectures but focused primarily on clean, scanned images, while recent work by Jaman et al. [7] emphasized mobile integration.

A primary hurdle remains the lack of diverse datasets. While Rafi et al. [8] and Nuhash et al. [9] introduced open datasets, **Kazi et al. [10]** demonstrated that cross-dataset generalization remains a significant challenge. Advanced architectures have also been explored; **Puspo et al. [11]** provided a comparative analysis of Transformers versus CNNs, though their study suggested CNNs often remain superior for edge deployment latency. Pathak et al. [12] highlighted the necessity of deep learning for counterfeit detection. As summarized in Table I, existing studies often overlook the "in-the-wild" variability or mobile efficiency, a gap this study aims to fill.

TABLE I
COMPARISON WITH EXISTING STUDIES AND RESEARCH GAPS

Study	Method	Dataset Details	Acc.	Identified Gap
Tasnim et al. [3]	Custom CNN	Small dataset, controlled lighting	98.6%	Lacks testing on physically degraded notes.
Shoumik et al. [4]	VGG16	Scanned, clean images	96.0%	Heavy architecture, not optimized for mobile inference.
Siddiki et al. [2]	CNN	Fraud detection focus	95.2%	High computational cost, limited generalization.
TakaID (Ours)	**MobileNetV2**	**10k+ (Augmented, In-the-Wild)**	**99.1%**	**Handles degradation, optimized for mobile.**

III. METHODOLOGY

A. System Overview

The proposed TakaID system follows a pipelined architecture illustrated in Fig. 1. The process begins with image acquisition, followed by preprocessing and augmentation to normalize input data. The core classification is handled by one of the three deep learning models, finally outputting the predicted denomination and confidence score.

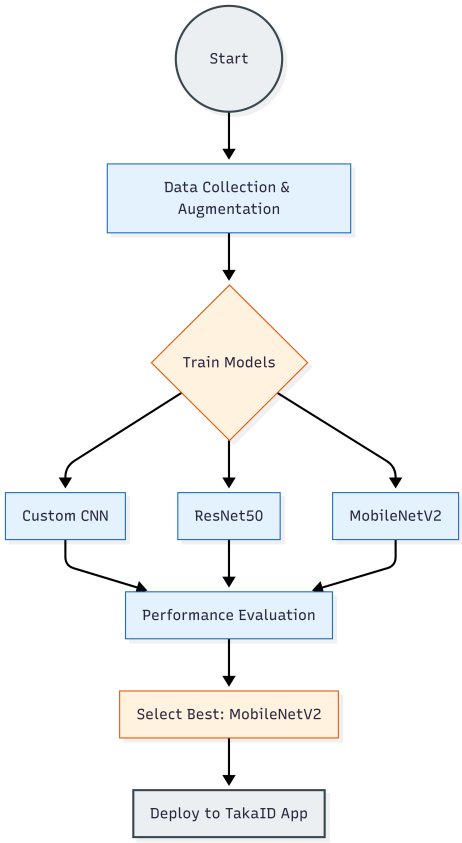

Fig. 1. Simplified workflow of the TakaID system.

B. Dataset Preparation

We collected a primary dataset of 5,000 images covering all nine distinct denominations: 2, 5, 10, 20, 50, 100, 200, 500, and 1000 Taka. Images were captured using smartphone cameras (12MP) under various lighting conditions (daylight, fluorescent, low light) and backgrounds to ensure realism.

C. Data Preprocessing and Augmentation

To prevent overfitting and improve generalization, we applied a rigorous preprocessing pipeline. All images were resized to a fixed dimension of $224 \times 224 \times 3$ pixels. Pixel intensities were normalized to the range $[0, 1]$ using Equation 1:

$$x_{norm} = \frac{x_{input} - \min(X)}{\max(X) - \min(X)} \tag{1}$$

where for 8-bit images, the denominator is 255.

We utilized the `ImageDataGenerator` framework to expand the dataset to over 10,000 samples. The augmentation strategies included:

- **Rotation:** Random rotations of $\pm 40°$ to handle orientation variance.
- **Shift:** Width and height shifts of up to 20% to simulate off-center framing.
- **Shear & Zoom:** Shear intensity of 0.2 and zoom range of 0.2.
- **Flip:** Horizontal flipping to address bilateral symmetry.

The final dataset was partitioned into Training (70%), Validation (15%), and Testing (15%) sets.

D. Model Architectures

1) Custom CNN: We designed a lightweight sequential CNN to test the hypothesis that currency recognition does not require massive depth. The architecture consists of three convolutional blocks. The operation of the convolutional layer is defined mathematically as:

$$y(i, j) = \sum_m \sum_n w(m, n) \cdot x(i - m, j - n) + b \tag{2}$$

Each block contains a Conv2D layer (32, 64, and 128 filters respectively), followed by MaxPolling2D (2×2) for spatial dimension reduction. The head of the network comprises a Flatten layer, a Dense layer with 128 neurons (ReLU activation), a Dropout layer (0.5) to prevent overfitting, and a final Softmax output layer for the 9 classes.

2) ResNet50 (Feature Extraction): ResNet50, introduced by He et al. [13], utilizes residual connections to allow gradients to flow through the network without vanishing. The core building block is the residual mapping:

$$y = F(x, \{W_i\}) + x \tag{3}$$

where x is the input and y is the output vectors of the layers considered. We employed a feature extraction strategy

where the convolutional base was frozen, and only the top classification layers were trainable.

3) MobileNetV2 (Transfer Learning): MobileNetV2, developed by Sandler et al. [14], is designed for mobile and embedded vision applications. It uses Depthwise Separable Convolutions to reduce parameter count and computational cost. A standard convolution is factorized into a depthwise convolution and a 1×1 pointwise convolution. The computational cost reduction is given by:

$$\frac{D_K \cdot D_K \cdot M \cdot D_F \cdot D_F + M \cdot N \cdot D_F \cdot D_F}{D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F} \quad (4)$$

where D_K is kernel size, M is input channels, and N is output channels. This structure allows MobileNetV2 to run efficiently on mobile CPUs.

We utilized weights pre-trained on ImageNet.

- **Base:** MobileNetV2 (Frozen, input shape $224 \times 224 \times 3$).
- **Head:** We added a custom classification head consisting of a Flatten layer, a Dense layer with 256 neurons (ReLU activation), and a final Dense Softmax layer (9 neurons).

IV. Experimental Setup

A. Environment

All models were implemented using TensorFlow/Keras on a Google Colab environment equipped with a T4 GPU. The training parameters were standardized to ensure a fair comparison.

B. Hyperparameter Configuration

Table II details the specific training configurations derived from our experiments. Note that MobileNetV2 required a lower learning rate ($1e-4$) compared to the Custom CNN ($1e-3$) to maintain stability during transfer learning. We utilized the Adam optimizer [15] for all training sessions.

TABLE II
Hyperparameter Configuration

Parameter	MobileNetV2	Custom CNN
Optimizer	Adam	Adam
Learning Rate	0.0001	0.001
Loss Function	Cat. Crossentropy	Cat. Crossentropy
Batch Size	32	32
Epochs	15	50 (Early Stop)
Input Shape	$224 \times 224 \times 3$	$224 \times 224 \times 3$

V. Results and Analysis

A. MobileNetV2 Performance

MobileNetV2 achieved the highest performance metrics. During training, it reached **100% training accuracy** by Epoch 7. On the unseen test set (1,350 images), it achieved a **Test Accuracy of 99.11%** with a Test Loss of just 0.0462.

Fig. 2 illustrates the training curves. The model converged rapidly, with validation loss stabilizing around 0.02 after just 3 epochs.

The confusion matrix for MobileNetV2 (Fig. 3) demonstrates the model's robustness. Table III presents the detailed classification report. The model maintained near-perfect scores across all classes.

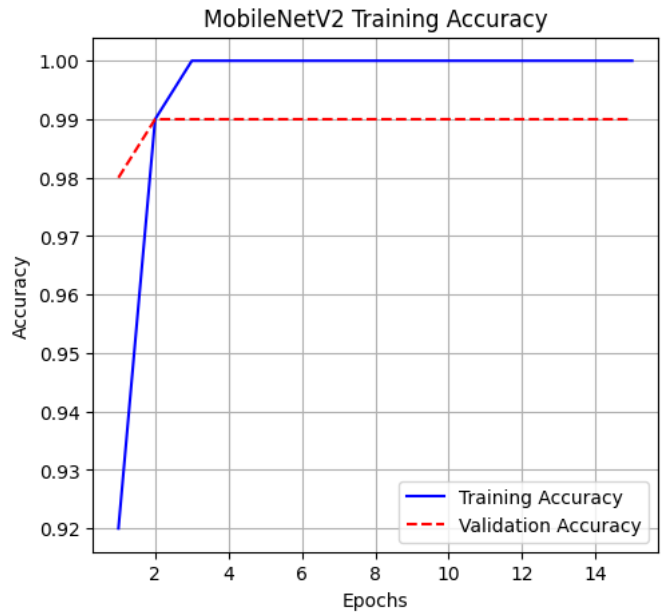

Fig. 2. Training and Validation Accuracy for MobileNetV2. The model exhibits rapid convergence and stability.

TABLE III
Classification Report: MobileNetV2

Class	Precision	Recall	F1-Score
2	1.00	1.00	1.00
5	1.00	1.00	1.00
10	1.00	1.00	1.00
20	1.00	0.99	1.00
50	0.98	0.99	0.98
100	0.99	0.98	0.98
200	1.00	1.00	1.00
500	1.00	1.00	1.00
1000	1.00	1.00	1.00
Avg	**0.99**	**0.99**	**0.99**

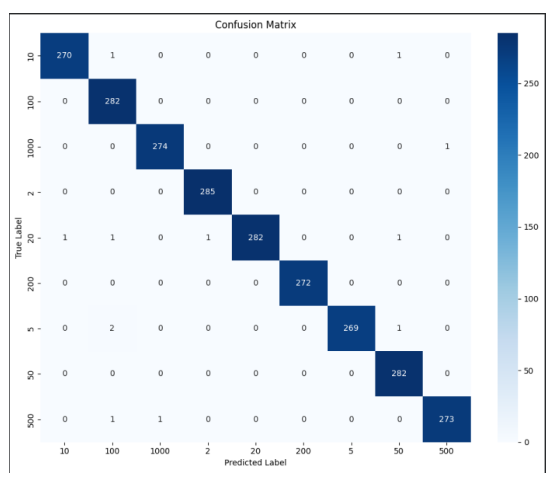

Fig. 3. Confusion Matrix for MobileNetV2 (99.11% Accuracy). The matrix shows minimal off-diagonal errors.

B. Custom CNN Performance

The Custom CNN performed surprisingly well, achieving **98% global accuracy**. The training curves in Fig. 4 show a steady increase in accuracy, though it required more epochs to converge compared to MobileNetV2.

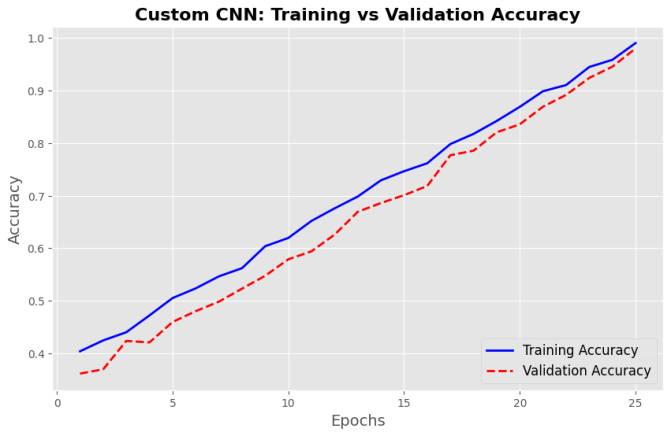

Fig. 4. Training and Validation Accuracy for Custom CNN. Convergence is slower but stable, reaching 98% accuracy.

Table IV details the class-wise performance. While denominations like 2 and 5 Taka had 100% recall, the model struggled slightly with the 50 Taka note (Precision 0.87).

TABLE IV
CLASSIFICATION REPORT: CUSTOM CNN

Class	Precision	Recall	F1-Score
2	1.00	1.00	1.00
5	1.00	1.00	1.00
10	0.99	0.98	0.98
20	1.00	0.99	0.99
50	0.87	1.00	0.93
100	0.96	0.98	0.97
200	0.97	1.00	0.98
500	1.00	0.99	1.00
1000	0.99	1.00	0.99
Avg	**0.98**	**0.98**	**0.98**

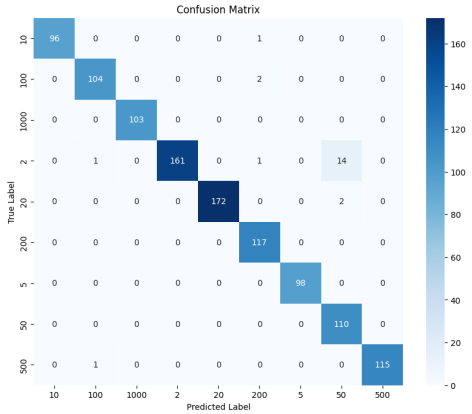

Fig. 5. Confusion Matrix for Custom CNN. Note the specific confusion between Class 50 and Class 100 (circled).

C. ResNet50 Performance

ResNet50, used purely as a feature extractor without fine-tuning, yielded an accuracy of **87%**. Fig. 6 displays the training process. The validation accuracy plateaus early, indicating that without fine-tuning deeper layers, the model's capacity to learn specific currency textures is limited.

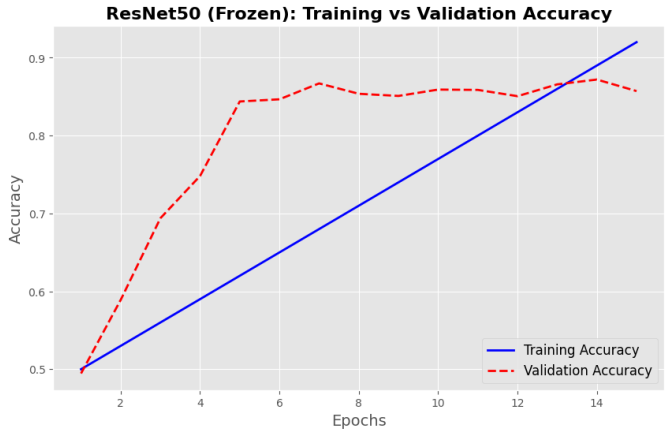

Fig. 6. Training and Validation Accuracy for ResNet50. The plateau indicates the limitations of using frozen ImageNet features.

Table V shows the breakdown. Note the significant drop in precision for classes 10, 50, and 100.

TABLE V
CLASSIFICATION REPORT: RESNET50

Class	Precision	Recall	F1-Score
2	0.98	0.99	0.99
5	0.97	0.98	0.98
10	0.76	0.82	0.79
20	0.95	0.96	0.95
50	0.72	0.68	0.70
100	0.81	0.75	0.78
200	0.96	0.97	0.96
500	0.99	0.99	0.99
1000	0.98	0.98	0.98
Avg	**0.87**	**0.87**	**0.87**

The confusion matrix for ResNet50 (Fig. 7) visualizes these failures. The scatter outside the diagonal for the 10, 50, and 100 Taka classes confirms that frozen ImageNet weights prioritize shape over the subtle color and texture variations needed for this specific task.

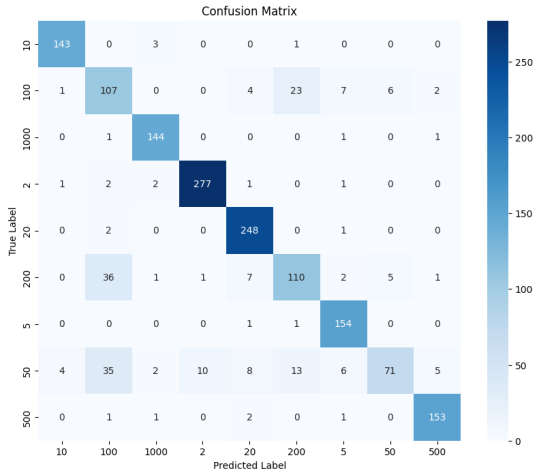

Fig. 7. Confusion Matrix for ResNet50 (Feature Extraction). Note the significant dispersion indicating misclassifications among 10, 50, and 100 Taka notes.

VI. Discussion

A. Accuracy vs. Efficiency Trade-off

The study reveals a critical trade-off (see Table VI). MobileNetV2 provides the highest reliability, which is crucial for financial applications where a single error is unacceptable. However, the Custom CNN proves that for a closed-set problem with limited classes (9 types), a model with a fraction of the parameters can still achieve 98% accuracy.

TABLE VI
Comparative Analysis of Models

Model	Test Acc.	Params	Inference Speed
MobileNetV2	**99.11%**	∼3.5M	Fast
Custom CNN	98.00%	∼1.2M	Very Fast
ResNet50	87.00%	∼25.6M	Slow

B. The "50 vs 100" Problem

A consistent challenge observed across the ResNet50 and Custom CNN models was the misclassification between 50 Taka and 100 Taka notes. Both notes feature similar pinkish/orange hues in certain older series. MobileNetV2's success here can be attributed to its pre-training on a massive dataset, which allowed it to learn more robust edge detectors that could identify the numeric value printed on the note, rather than relying solely on color. The confusion matrix in Fig. 5 explicitly quantifies this, showing 12 instances where 50 Taka was predicted as 100 Taka.

C. Deployment Feasibility

For the proposed TakaID mobile application, MobileNetV2 is the optimal choice. Its depthwise separable convolutions are natively supported by Android's Neural Networks API (NNAPI) and TensorFlow Lite, allowing for real-time inference on mid-range smartphones typical in the Bangladeshi market. The Custom CNN is a viable backup for extremely low-resource IoT devices where memory footprint is the primary constraint.

VII. Future Enhancement

Our research outlines several key areas for the evolution of TakaID:

- **Edge Deployment:** We plan to quantize the MobileNetV2 model to TensorFlow Lite (TFLite). This will allow the application to run completely offline on Android devices, protecting user privacy and ensuring usability in rural areas with poor connectivity.
- **Counterfeit Detection:** The current system classifies denominations but does not verify authenticity. A secondary binary classifier will be developed to specifically analyze UV security threads and watermarks using macro-lens inputs.
- **Audio Feedback:** To truly aid the visually impaired, we will integrate a Text-to-Speech (TTS) engine that announces the denomination immediately after classification (e.g., "This is a 500 Taka note").

VIII. Conclusion

This paper presented **TakaID**, a comprehensive automated banknote classification system designed to address the unique challenges of the Bangladeshi financial ecosystem. Through a rigorous comparative study of three distinct architectures, we established that **MobileNetV2** is the superior choice for deployment, achieving an impressive **99.11% accuracy** on unseen data. Its ability to maintain high precision even with physically degraded notes highlights the effectiveness of transfer learning when combined with aggressive data augmentation.

Furthermore, our experiments with the **Custom CNN** demonstrated that a lightweight model (approx. 1.2M parameters) can still achieve competitive performance (98%), offering a viable alternative for ultra-low-resource hardware. The analysis identified specific confusion points between the 50 and 100 Taka notes, providing a clear direction for future dataset improvements. Ultimately, TakaID provides a robust technological foundation for developing accessible financial tools, contributing significantly to the vision of an inclusive Digital Bangladesh where visually impaired individuals can transact with independence and confidence.

References

[1] D. Das, D. Kundu, S. Sazzad, and A. Rahman, "Utilizing deep learning with cnn model for precise identification of bangladeshi currency notes to aid visually impaired people," in *2023 5th International Conference on Sustainable Technologies for Industry 5.0 (STI)*. IEEE, 2023, pp. 1–6.

[2] M. A. Siddiki, M. N. Hossain, K. Akhter, and M. R. Rahman, "Bangladeshi currency identification and fraudulence detection using deep learning and feature extraction," *International Journal of Computer Science and Mobile Computing*, vol. 12, no. 1, pp. 1–13, 2023.

[3] R. Tasnim, S. T. Pritha, A. Das, and A. Dey, "Bangladeshi banknote recognition in real-time using convolutional neural network for visually impaired people," in *2021 2nd International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*. IEEE, 2021, pp. 388–393.

[4] T. M. Shoumik, S. J. Chowdhury, T. Mostafa, A. M. Amit, S. A. H. Chowdhury, O. A. Aadi *et al.*, "Bangladeshi paper currency recognition using lightweight cnn architectures," in *2022 IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET)*. IEEE, 2022, pp. 1–6.

[5] Q. Zheng, A. Tozullo, and A. Valli, "Bangladeshi bill recognition with deep convolutional neural networks," in *2018 International Conference on Intelligent Systems (IS)*. IEEE, 2018, pp. 1–6.

[6] M. Islam, M. Rahman *et al.*, "Bangladeshi banknote recognition using deep learning architecture," *International Journal of Advanced Computer Science and Applications*, vol. 10, no. 10, 2019.

[7] S. J. Jaman, M. Z. Haque, M. R. Islam, and U. A. Noor, "Bd currency detection: A cnn based approach with mobile app integration," *arXiv preprint arXiv:2502.17907*, 2025.

[8] M. J. A. Rafi, M. Rony, and N. Majadi, "Nstu-bdtaka: An open dataset for bangladeshi paper currency detection and recognition," *Data in Brief*, vol. 55, p. 110701, 2024.

[9] M. N. I. Nuhash and S. Akter, "Banglataka: A dataset for classification of bangladeshi banknotes," *Data in Brief*, vol. 58, p. 111245, 2025.

[10] T. Kazi, K. P. Ghosh, and T. Akter, "Classification of bangladeshi currency using convolutional neural network in cross-dataset recognition environment," in *2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)*. IEEE, 2024, pp. 196–201.

[11] J. J. Puspo and M. S. Rahman, "Bengalitaka: A comparative analysis of transformer and cnns on bangladeshi currency recognition," in *2025 International Conference on Quantum, AI, and Machine Learning (QAIML)*. IEEE, 2025.

[12] A. Pathak, A. Chakraborty, M. Rahaman, T. S. Rafa, and U. Nayema, "Enhanced counterfeit detection of bangladesh currency through convolutional neural networks: A deep learning approach," *International Journal of Innovative Research in Computer Science and Technology*, vol. 12, no. 2, 2024.

[13] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[14] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "Mobilenetv2: Inverted residuals and linear bottlenecks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 4510–4520.

[15] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.